Earth and Planetary Science Letters, 44 (1979) 176–192
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

ASPECTS OF CALEDONIAN PALAEOMAGNETISM AND THEIR TECTONIC IMPLICATIONS

J.D.A. PIPER

Sub-Department of Geophysics, University of Liverpool, Liverpool L69 3BX (U.K.)

Received September 11, 1978
Revised version received February 28, 1979

Palaeomagnetic results from the Lower Palaeozoic inliers of northern England cover the upper part of the (Middle Ordovician) Borrowdale Volcanic Series (palaeomagnetic pole 208°E, 18°S, A_{95} = 9.4°), minor extrusive units relating to the Caradoc and Ashgill stages of Ordovician times, intrusive episodes of Middle Ordovician and Middle Silurian to Late Devonian age, and the Shap Granite of Devonian (393 m.y.) age (palaeomagnetic pole 313°E, 33°S, A_{95} = 5.6°).

A complete assessment of Ordovician to Devonian palaeomagnetic data for the British region shows that the pole was nearly static relative to this region for long intervals which were separated by shifts occupying no more than a few millions of years. The mean palaeomagnetic poles are: Ordovician (6°E, 16°S), Lower Silurian (58°E, 16°N), Middle Silurian/Lower Devonian (318°E, 6°N) and Middle/Upper Devonian (338°E, 26°S); the first two shifts separating these mean poles can be explained predominantly in terms of rotational movements of the crustal plate but the last involved appreciable movement in palaeolatitude.

Comparison of Lower Palaeozoic palaeomagnetic data from the British region with contemporaneous data from continental Europe/North America on the Pangaean reconstruction reveals a systematic discrepancy in palaeolatitude between the two regions prior to Middle Devonian times. This discrepancy was eliminated during a few millions of years of Lower/Middle Devonian times (ca. 395 m.y.) and can be explained in terms of ca. 3500 km of sinistral strike-slip movement close to the line of the orthotectonic Caledonides. This motion is linked both in time and place to the impingement of the Gondwanaland and Laurentian supercontinents during the Acadian orogeny; this appears to have displaced the British sub-plate until it became effectively locked between the Baltic and Laurentian regions. Although movement of the dipole field relative to the British region in Lower Palaeozoic times is now well defined, nearly one fifth of the total data show that the geomagnetic field was more complex than dipolar during this interval. Until the significance of these anomalies is fully resolved, the tectonic model derived from the palaeomagnetic data cannot be regarded as unambiguous.

1. Introduction

The past decade has seen recognition [1,2] and extensive discussion of the Iapetus (Proto-Atlantic) suture, an episode of ocean basin formation and consumption intimately related to the emplacement and folding of the Palaeozoic Caledonian/Appalachian orogenic belt. This discussion complements earlier debate into major thrust and strike-slip movements within the belt [3]. The resulting models link several facets of the orogeny [1,4–7] and involve palaeogeographic reconstructions that need to be tested against the palaeomagnetic data. Although these data cannot distinguish between the fine detail of the models, they do impose constraints on their gross predictions; they also have several implications which must in turn be tested against the geological evidence.

This paper describes the palaeomagnetism of Palaeozoic inliers in northern England and examines their implications to the definition of apparent polar wander (a.p.w.) movements during the interval of the Caledonian orogeny. A comparison is also made with contemporaneous results for the regions of Continental Europe and North America which adjoin the British region on the Pangaean reconstruction.

2. Palaeomagnetism of Palaeozoic inliers, northern England

The major Lower Palaeozoic igneous episode in the north of England is represented by the Borrowdale Volcanic Series which occupies the larger part of the Lake District inlier (Fig. 1). It comprises a pile of intermediate to acid lavas and tuffs erupted between the late Llanvirn and Lower Caradoc stages of Ordovician time [8] and locally reaches a thickness of 7000 m. Palaeomagnetic results were reported by Morris [9] and Faller et al. [10] and these are combined in Table 2 with four additional sites reported by Piper et al. [11]; with the exception of three sites [9], this work was concentrated on the lower part of the volcanic succession. Additional work will be necessary to evaluate the magnetic stratigraphy and the significance of the high percentage of anomalous directions found by the earlier workers.

The upper part of the Borrowdale Volcanic Series has accordingly been sampled where detailed structural and stratigraphic control is available (based on the field studies of Mitchell [12]). The succession comprises approximately 1000 m of massive acid and intermediate lavas and their tuffaceous equivalents (Fig. 1) between units classified as the Harter Fell Rhyolites and the Upper Rhyolites; structural dips recorded by Mitchell [12] and during the present study from tuffaceous horizons range from 32° to 60° in a south-southeast direction.

All samples were progressively cleaned by 500 c/s a.f. demagnetisation to peak fields ranging from 30 to 120 mT; selected samples were also thermally cleaned. The volcanics possess medium (5–100 mT) coercive force (c.f.) components and move rapidly to stable end points (Fig. 2); less stable components were present at sites 10 and 14. Thermomagnetic determinations (saturation magnetisation, J_s, versus

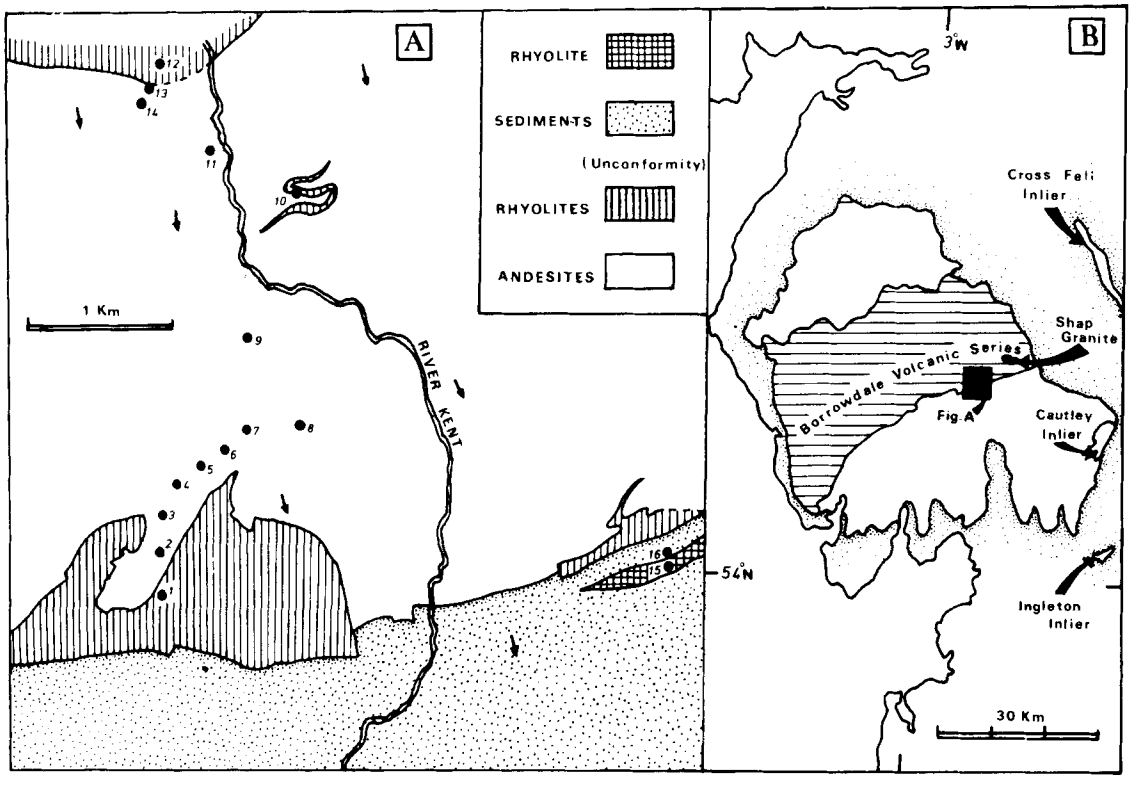


Fig. 1. A. Simplified geology and sampling sites of the upper part of the Borrowdale Volcanic Series and the post-Borrowdale rocks in the Kentmere area [12], English Lake District; the arrows indicate directions of structural dip. B. Outline map of northern England showing the location of the sampled area of A in relationship to the total outcrop of the Borrowdale Volcanic Series, together with the Lower Palaeozoic inliers for which palaeomagnetic data are listed in Tables 1 and 2.

TABLE 1

Palaeomagnetic results from Palaeozoic rocks of Northern England after a.f. and thermal cleaning

Site No.	N	D	I	k	α_{95}	Total NRM intensities (A m^2/kg)	Curie point(s) (±10°C)
(1) Borrowdale Volcanic Series (Upper Part):							
1	4	334.2	−24.2	52	12.9	1.65–0.48	560, C > H
2	6	317.3	−20.8	71	8.0	3.84–1.36	550, C > H
3	7	320.7	−31.8	52	8.5	1.41–0.57	520, C > H
4	3	334.7	−19.3	27	24.3	0.95–0.24	500, H = C
5	7	306.5	5.0	98	6.1	3.13–0.82	565, C > H
6 *	6	103.8	57.2	106	6.5	0.33–0.20	575, C > H
7	6	332.4	−18.5	53	9.3	0.83–0.09	550, C > H
8	7	327.8	−46.5	55	8.2	1.15–0.06	560, C > H
9	5	321.9	−33.2	79	8.7	2.29–1.02	540, C > H
10 *	7	262.0	11.5	9	21.9	16.60–3.02	570, H = C
11 *	3	254.6	38.0	15	33.5	0.07–0.02	570, C > H
12	5	327.3	−1.0	25	15.7	0.96–0.18	570, C > H
13	5	1.9	−31.0	19	18.2	0.57–0.09	580, C > H
14	7	354.0	−26.1	77	6.9	0.52–0.23	570, H = C
						Palaeomagnetic pole	
Mean of 12 sites: (inc. site 10E [10])		329.7	−25.3	15	11.4	208.2°E, 17.5°N (dp = 6.6°, dm = 12.3°)	
(2) Round Knott Dolerite (Carrock Fell Complex):							
15		184.1	29.5	16	10.3	15.50–0.10	540–575°
						Palaeomagnetic pole 352.8°E, 19.6°S (dp = 6.3°, dm = 11.5°)	
(3) Post-Borrowdale igneous activity at Kentmere:							
Stockdale Rhyolite:	8	169.5	−73.3	17	13.9	0.78–0.03	575°, C > H
						(VGP: 11°E, 44°S)	
Lamprophyre dyke:	8	357.3	−40.0	16	14.5	0.13–0.03	565°, H = C
(4) Cautley Fell Inlier:							
Cautley Rhyolite:	18	190.0	−30.2	8	12.8	2.44–0.02	***, H = C
						(VGP: 342°E, 51°S)	
Cautley Dolerite:	19	231.4	26.7	5	16.3	0.09–0.01	***, H = C
						(VGP: 308°E, 9°S)	
(5) Ingleton Inlier, lamprophyre dykes:							
Dyke 1	5	100.7	76.7	70	9.2	5.49–1.19	570°, H = C
Dyke 2	7	199.1	14.3	9	21.9	0.09–0.02	565°, H = C
(6) Shap Granite (393 m.y.):							
	21	216.5	−10.7	20	7.3	0.52–0.04	550–565° 660–670°, H = C
						Palaeomagnetic pole 312.6°E, 32.8°S (dp = 3.8°, dm = 7.4°)	

N is number of samples yielding combined mean direction D, inclination I and precision parameter k [$=(N-1)/(N-R)$], with 95% confidence circle radius α_{95} about mean direction; dp and dm are radii of semi-axes of 95% confidence oval about the palaeomagnetic pole in the colatitude direction at right angles to it respectively.

* Sites excluded from group mean calculation.

** H, C = J_S on heating, cooling.

*** Exponential decrease in J_S.

temperature) exhibit a sharp drop in saturation magnetisation at low temperatures (Fig. 2C) with a residual component yielding a Curie point close to pure magnetite; no hematite is observed. Site mean statistics are listed in Table 1 after correction for tectonic tilt; eleven sites have normal magnetisations, two (sites 10 and 11) have anomalous directions (>40° from the mean), and site 6 is close to (but not precisely) reversed. Site 6 is overlain by site 5 which is also slightly removed from the remaining sites and these are excluded from calculation of the mean. Morris [9] reports three sites from the stratigraphic level of sites 4–8 yielding one normal and two anomalous directions; the normal direction (site 10E) is comparable to this study and is accordingly included in the group mean.

Since the tilts of many of the sites in this collection are rather similar, a fold test is not significant at the 95% confidence level [13] but the Fisher precision parameter (k) improves from 10 to 15 when this correction is applied. The sequence of direction changes through the volcanic succession is illustrated in Fig. 2. The group mean direction (Table 1) is slightly less steep than the mean derived from the lower part of the Borrowdale succession [9–11] and nearer to the mean of Ordovician studies from elsewhere in Britain (Table 2); only minor a.p.w. movement can thus be identified during the interval of accumulation of the volcanic sequence.

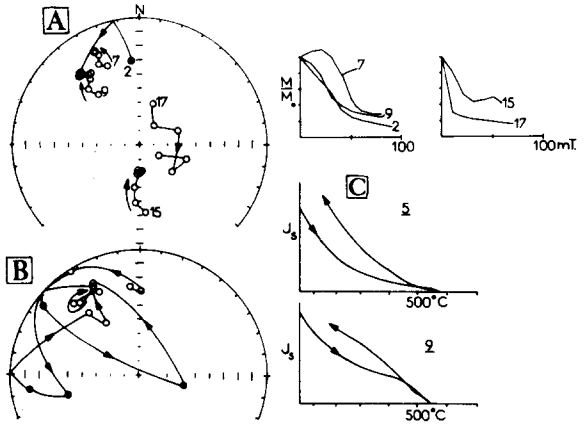


Fig. 2. A. Behaviour of directions of magnetisation and magnetic moments (expressed as a fraction of initial moment, M_0) of typical samples of the Borrowdale Volcanics with progressive a.f. treatment; the numbers refer to site numbers. B. The change in direction of magnetisation (after cleaning) with stratigraphic position through the top 1000 m of the Borrowdale succession at Kentmere. C. Typical thermomagnetic curves for these volcanics.

The Borrowdale volcanism was terminated by folding during the late Llandeilo or Lower Caradoc stages of Ordovician times [8] followed by deposition of sediments in mid-Caradoc (Actonian) times and residual volcanism [12] represented by the Stockdale Rhyolite (site 15, Fig. 1). The ⩾10-mT c.f. component of magnetisation present in this unit yields the mean listed in Table 1 after correction for Late Silurian folding. The virtual geomagnetic pole ([14], Table 2) is remote from other Ordovician poles but very similar to an anomalous palaeomagnetic pole (pole 15, Table 2) from the Cader Idris basalts of North Wales [14] also probably of Caradocian age. The Stockdale Rhyolite is cut by a lamprophyre dyke (site 16, Fig. 1) possessing a medium c.f. component of remanence residing in magnetite; the directional data are listed in Table 1 and by analogy with similar dykes in northern England [11] the dyke is inferred to be of Devonian age.

The Round Knott Dolerite is a 2-km^2 body intruding the Carrock Fell gabbro complex ([9,10], pole 10A, Table 2) at 357°E, 54.5°N; it predates sediments assigned to the Caradoc stage of Ordovician times and is thus of broadly comparable age to the Borrowdale Volcanic Series. A magnetite-held remanence with c.f. components in the range 10–80 mT is of reversed polarity (D = 184°, I = 30°, Table 1); two samples with higher c.f. spectra have a normal component of magnetisation (mean D = 346°, I = –5°), This body is thus largely of opposite polarity to the Carrock Fell gabbro which it intrudes, and to the Borrowdale Volcanic Series; the palaeomagnetic poles are similar.

The Lower Palaeozoic (Cautley) inlier at 357.5°E, 54.3°N (Fig. 1) includes a single rhyolite flow of middle/upper Ashgill age and an intrusive dolerite of post-Lower Silurian, pre-Carboniferous age [15,16]. Two outcrops of the rhyolite yield after cleaning a well-grouped site D = 186°, I = 20° (N = 8, k = 38) and a poorly-grouped site D = 195°, I = –40° (N = 10, k = 6); the mean for all 18 samples is listed in Table 1 and yields a virtual geomagnetic pole with similar longitude but southerly latitude with respect to other British Ordovician poles. The dolerite has an outcrop of 1 km^2 and lies close to the Dent fault [15], part of

a major Palaeozoic fault system in Northern England); this fault is probably related to the extensive hydrothermal alteration of the body. Stable, low (5–20 mT) c.f. components of magnetic remanence were recovered from 19 cores and lie close to the Upper Silurian/Lower Devonian field axis for Britain; 7 samples have normal and 12 reversed magnetisations with the mean (assuming dual polarity) listed in Table 1.

The Lower Palaeozoic (Ingleton) inlier at 357.7°E, 54.0°N (Fig. 1) includes two lamprophyre dykes of post-Llanvirn and pre-Carboniferous age. One is a west-northwest-trending kersantite with mean remanence direction $D = 101°$, $I = +77°$ (Table 1); this is comparable to kersantite dykes with similar trend comprising a swarm within the Cross Fell Inlier 60 km to the north ([11], and Table 2, footnotes). A second impersistent lamprophyre has lower c.f. remanence of contrasting direction (Table 1) comparable to Middle/Upper Devonian main field directions.

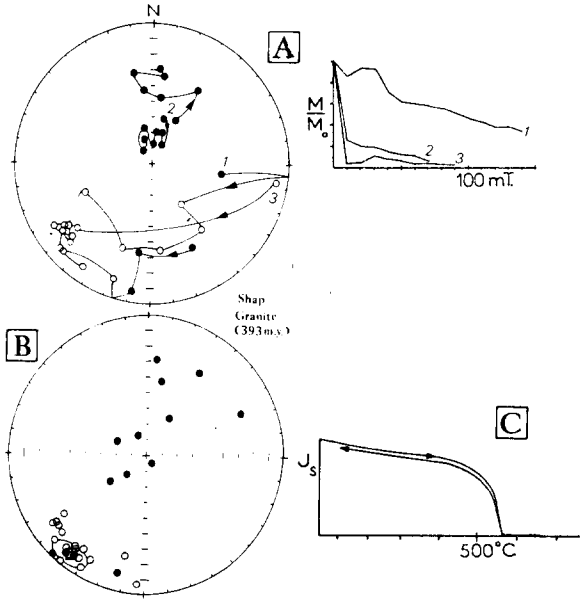


Fig. 3. A. Direction behaviour of some samples of the Shap Granite with progressive a.f. cleaning treatment. B. Cleaned sample directions of magnetisation for the granite together with the overall mean direction of magnetisation and 95% confidence circle for the reversely magnetised samples. C. A typical thermomagnetic determination from this granite.

The Shap Granite is a 5-km^2 intrusion at 357.3°E, 54.5°N; it was intruded and unroofed in Devonian times and the mean of many K-Ar and Rb-Sr age determinations is 393 m.y. [25]. Cryptocrystalline hematite gives a ubiquitous red colouration to this rock and is extensively visible in polished sections; thermomagnetic determinations in general exhibit single Curie points in the range 550–560°C. Although some show a tail attributable to hematite (Fig. 3C) residing in about 1% of the initial J_s. It is inferred that the bulk of the total NRM resides in large titanomagnetite grains (visible in polished sections) because a.f. demagnetisation produced progressive fall of NRM; coercivities however, are very variable and stable southeast directions of magnetisation (Fig. 3A) reside in moments with c.f. ranges of 5–30 mT in some samples and >100 mT in others. Six samples retained steep positive directions stable to a.f. and thermal treatment, and four samples yielded northeast directions with shallower inclinations (Fig. 3B); these subsidiary groups suggest that a reversal of magnetisation may have taken place during the cooling interval. The remaining 21 samples yield a group mean direction of $D = 217°$, $I = -11°$ ($\alpha_{95} = 7°$, Table 1).

The palaeomagnetic pole for the granite falls close to a group of poles of Middle-Upper Devonian age (Fig. 4), and this agreement improves when an adjustment is made for the tilt of Carboniferous sediments exposed 2 km to the east of the granite outcrop (Table 2, footnotes). Minette dykes spatially associated with, and directed towards, the granite [11] yield a similar pole ([38], and footnotes, Table 2).

3. Caledonian palaeomagnetic data for Britain

These new results are summarised in Table 2 together with other palaeomagnetic data (Ordovician/Devonian) relevant to the interval of the Caledonian orogeny; the results are plotted as palaeomagnetic *south* poles in Fig. 4. Most of the rock units yielding these data are variably deformed, sometimes by more than one episode of folding, and uncertainties remain in some of the results [17,19]; this problem relates mainly to Ordovician rocks which have experienced tectonism both in late Ordovician/early Silurian times and again in late Silurian/lower Devonian times [26],

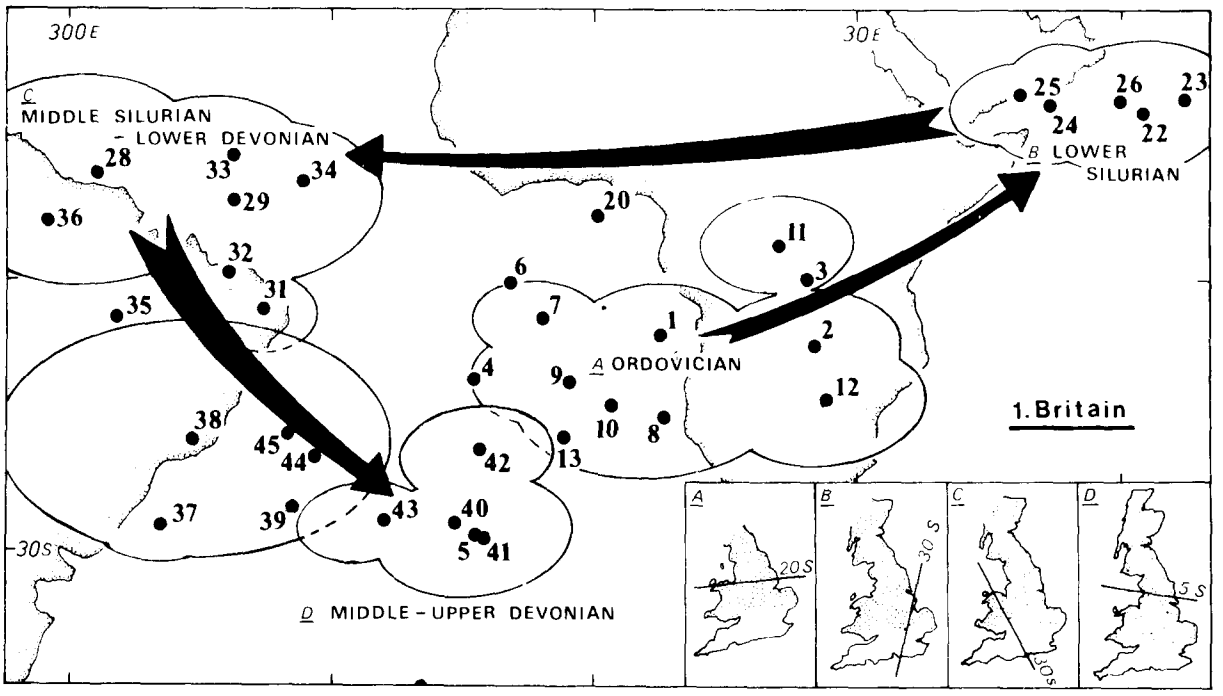


Fig. 4. A. Palaeomagnetic poles (Table 2) defining the apparent polar wander path for Britain from Ordovician to Upper Devonian times. The fields of 95% confidence ovals are shown for all results except those which cannot be considered to have averaged secular variation (Table 2) and the inset figures illustrate the areas to which the data from groups of poles *A* to *D* apply (stippled) with mean lines of palaeolatitude. Poles excluded from this figure are poles 14–19 and 21 the significance of which is discussed in the text. The quality of the pole determinations is outlined in Table 2 and the footnotes.

and probably contributes to the overall scatter of the Ordovician results in Fig. 4. However, all the poles listed in the table, except pole 42 which is corroborated by contemporaneous data, are defined by extensive a.f. demagnetisation tests and many (see table) by thermal demagnetisation and field tests as well. Even where a field test is not directly available, the convergence of separate poles from rocks of the same age after individual tilt corrections confirms that the magnetisations are penecontemporaneous and do not relate to widespread remagnetisation at later times [9,17].

The locations of the palaeomagnetic studies are illustrated in Fig. 5 with respect to the major (pre-Mesozoic) tectonic subdivisions of Britain. In the discussion that follows no selection of the data is made because this should logically be accompanied by a selection based on the age control; also such a selection would not affect the definition of the mean poles or the timing of the movements between them, and the remaining conclusions are drawn only from these overall trends in the data.

Table 2 includes data from both sides of the Iapetus suture (Fig. 5) a zone of subduction and obduction closely associated with folding and metamorphism of the paratectonic and orthotectonic Caledonides [6]. It is believed to have reached its maximum development in Lower Ordovician times (e.g. [4,6]) and comparisons between the Lower Ordovician palaeomagnetic data from either side of this suture have already been made [17,20]. They indicate an amount of post-Lower Ordovician closure of 1000 km or less because there is a broad overlap of confidence ovals of poles from either side of the suture (Fig. 6B); this figure is similar to structural and tectonic estimates for the amount of contraction across the paratectonic Caledonides [6,26]. The palaeomagnetic data are not very sensitive to strike-slip motion along the suture because the margins are oriented close to lines of palaeolatitude; however, Lower Ordovician magnetic

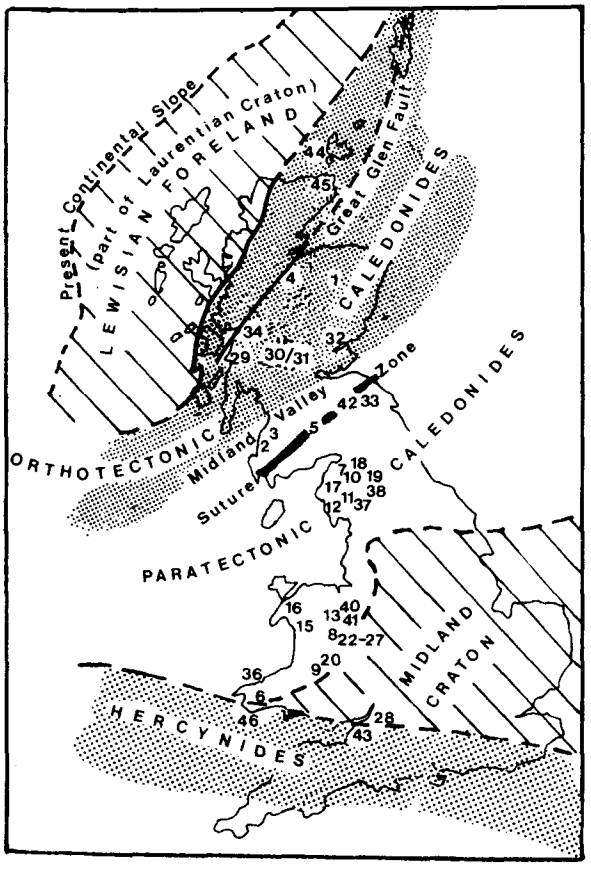


Fig. 5. The locations of the palaeomagnetic results of Fig. 4 and Table 2 within the pre-Mesozoic tectonic framework of Britain.

declinations are similar on both sides of the suture and show that rotational movements, at least, were minimal. The timing of final closure is not known, but this zone did not form a major faunal barrier after late Ordovician times and closure may well have been diachronous [6]. It is noteworthy that although there is no significant difference in the latitudes of the Middle Silurian/Lower Devonian palaeomagnetic poles from the north (poles 29–32 and 34, Fig. 4) and the south (poles 28, 33, 35, 36, Fig. 4) of the suture zone, there is a slight difference in the longitudes of these groups of poles (Fig. 4); this is consistent with (but cannot establish) the 1000 km of post-Lower Devonian dextral shear along the line of the suture postulated by Phillips et al. [6] to accommodate evidence for subduction continuing into Lower Old Red Sandstone times.

Two new facets of these data have recently complicated our understanding of a.p.w. movements during Lower Palaeozoic times. Firstly, a wide range of rock units from North Wales and the Northern Lake District (represented by poles 14–19 and at least in part of Upper Ordovician age) have been found to possess directions of magnetisation with steep inclinations [11,14] defined by extensive laboratory and limited field tests, and quite unlike the shallow directions of magnetisation hitherto regarded as typical of Britain in Lower Palaeozoic times [17]; discussion of these data is pursued below.

Secondly, rock units belonging to the late Ashgill/late Llandovery divisions of Upper Ordovician and Lower Silurian times yield poles 22–26 (Fig. 4) including at least one, and probably two, field reversals. The pole position (ca. 60°E, 15°N) has not yet been identified outside of the Welsh Borderland region (ca. 357°E, 525°N). It is, however, quite certain that these poles are not to be explained in terms either of local rotation of the sampled area or of anomalous geomagnetic field behaviour for three reasons: firstly, older rocks from the same area yield poles (numbers 8, 9, 13, Table 2) similar to contemporaneous rocks from elsewhere in Britain; secondly, magnetic inclinations and hence palaeolatitudes are comparable to other British Palaeozoic results; and thirdly, the sampled area is not separated by any major tectonic discontinuity from the remaining results from England and Wales ([19], and Fig. 5).

The collective data define four time groups of poles (*A* to *D* in Fig. 4) and each one includes poles of normal and reversed polarity. There is also a high degree of internal consistency in these data and, with the exception of two examples of later remagnetisation (poles 36 and 46, Table 2), only units 5 and 21 give poles falling outside of the main group relating to their age and in a group of different age. Indeed, the only overlap of confidence ovals is a marginal one between the Ordovician (*A*) and Middle/Upper Devonian (*D*) groups, but even here there is no overlap between the results based on large collections and the mean directions computed from the combined results for these intervals are quite different from one another; thus even if the extensive field and laboratory tests relating to these studies (Table 2) were to be ignored, this observation is a strong case against Devonian or later remagnetisation.

These time groupings of poles *A*–*D* are separated by rapid shifts which represent changes predominantly in declination and could therefore be largely accounted for by rotations of the British region (Fig. 4, insets) about a local axis. Also, the timing of these shifts can now be defined by results with good stratigraphic control (Table 2): thus shift from *A* to *B* took place during latest Ashgill or earliest Llandovery time and the shift from *B* to *C* is restricted by the stratigraphic ages of the poles 22–26 and pole 28 to the latest Llandovery or Lower Wenlock stages of Silurian times.

The shift from group *C* to group *D* took place close to the Lower/Middle Devonian boundary: 32–34 falling in group C are of Lower Devonian age while 39 and 42–45 falling in group *D* are of Middle and Upper Devonian age; poles 40 and 41 falling in group *D* cannot be assigned to a specific division of Devonian times [19]. The Midland Valley lavas and the Cheviot lavas of group *C* (poles 32 and 33) are dated ca. 397 and 390 m.y. respectively, while the Shap Granite of group *D* is dated 393 m.y. Absolute age data relating to Palaeozoic times [27] define the Devonian period at 415–360 m.y. (employing the same Rb-Sr decay constant as in Table 2) and the Lower/Middle Devonian transition between 400 and 380 m.y.; these limits embrace the radiometric evidence for the timing of the shift $C \rightarrow D$ and are independently confirmed by the stratigraphic ages.

Unfortunately, most of the Lower Palaeozoic palaeomagnetic results from Ireland [28,29] are only of a reconnaissance nature or are associated with uncertain tectonic corrections, and little useful comparison can be made with the data from Britain at this stage. Ordovician results (Fig. 6A) are distinct from the British Ordovician data in magnetic declination and Morris [29] interprets this in terms of regional rotation of the South Mayo Trough. Middle Silurian (Wenlock) results from Ireland (6 and 7 in Fig. 6A) lie in the fields of the post-Lower Silurian poles from Britain and are consistent with the shift $B \rightarrow C$ in Fig. 4, although agreement between these two poles is poor.

Thus the a.p.w. path for the British region from Ordovician to Devonian times is defined by nearly static positions of the pole for long periods of time (*A*, *B*, *C* and *D* in Fig. 4) separated by rapid shifts completed during intervals of no more than a few mil-

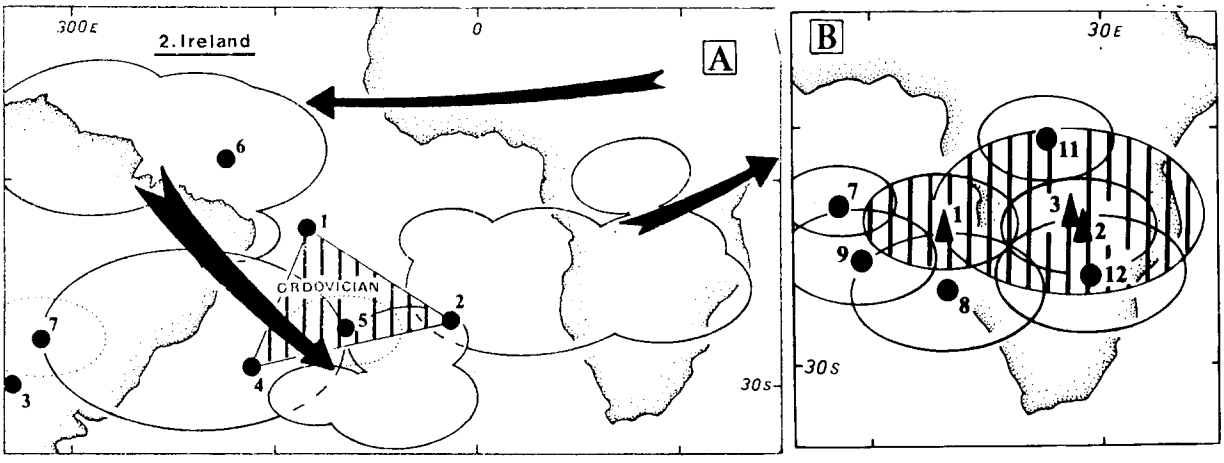


Fig. 6. A. Lower Palaeozoic palaeomagnetic results from Ireland [28,29] compared with the distribution of contemporaneous poles from Britain. Poles: 1 = Glensoul Felsite (post-Arenig); 2 = Kildare inlier (pre-Ashgill); 3 = Balbriggen Volcanics (Ordovician); 4 = Portrane Volcanics (Ordovician); 5 = Grangegeeth Volcanics (Ordovician); 6 = Knocknaveen Group (Wenlock); 7 = Salrock Group (middle/upper Wenlock). Results 1, 6 and 7 are from the north and results 2–5 from the south of the Iapetus suture in Ireland [6]. Data from the Connemara gabbros and Mweelrae ignimbrites are excluded because of tectonic complexities. Confidence ovals are not included where these are large or the results are based on few samples only. The shaded triangle embraces the Ordovician results with the exception of pole 3. B. Lower/Middle Ordovician palaeomagnetic poles with 95% confidence ovals from the north (triangles with ovals shaded) and south (circles) sides of the Iapetus suture in Britain (Table 2 and Fig. 5).

TABLE 2

A compilation of Ordovician to Devonian palaeomagnetic results from Britain

Pole no.	Rock unit	Radiometric age (m.y.)	Stratigraphic age	Polarity (field and lab. analysis)	Palaeo-magnetic pole	*dp/dm*
(1) Ordovician rocks, northern margin of Iapetus suture						
1	Younger Gabbros Aberdeenshire	486–461 *,**	–	R(a)''''	9E, 11S	6/10
2	Ballantrae Complex, Serpentinites	475 **	LN-LD	R(a, t)''''	27E, 12S	6/10
3	Byne Hill Gabbro	–	LN-LD	R(a, t)'''	26E, 10S	11/19
4	Foyers Complex [1]	420–460 **	.	R(a)'''	348E, 16S	12/21
5	Bail Hill Volcanics	445 **	C_l	N(a)''	168E, 33N	6/12
(2) Ordovician rocks, southern margin of Iapetus suture						
6	Trefgarn Andesitic Series		AR	R(a)'''	352E, 5S	19/27
7	Eycott Group Lavas		LN_l	N(a, f)''''	176E, 9N	4/ 7
8	Stapley Hill Volcanic Group		LN_l	M(a, f)'''	10E, 20S	7/13
9	Builth Volcanic Series		LN_{l-m}	R(a)''''	359E, 16S	6/11
10A	Carrock Fell Complex [12]		>C	N(a)''''	184E, 19N	6/11
10B	Round Knott Dolerite		<10A, >C	R(a)''	353E, 20S	6/12
11	Borrowdale Volcanic Group, A [2]		LN-LD	R(a, f)''''	23E, 1S	6/ 9
12	Borrowdale Volcanic Group, B [3]		>C	R(a, t)''''	28E, 18S	7/12
13	Tuffs, Welsh Borderlands	451 ***	C	R(a, f)'''	358E, 23S	11/20
14	Stockdale Rhyolite [4]		C	(a)'	130E, 83S	–
15	Cader Idris Basalts		~C	–(a)'''	229E, 73S	18/22
16	North Wales Igneous Province [5]		–	–(a)''''	108E, 68S	8/ 9
17	Threlkeld–St. Johns Microgranodiorite	458 *	–	–(a)'''	88E, 68N	20/25
18	WNW Dolerite Dykes, Carrock Fell		<LN	–(a)'''	124E, 68N	13/17
19	WNW Kersantite Dykes, Cross Fell [6]		<LN	–(a)'''	337E, 54N	26/26
20	Dolerite Intrusions, Builth Inlier		C_u-AS_l	M(a)'''	2E, 2N	11/16
21	Cautley Fell Rhyolite [7]		AS_{m-u}	R(a, t, f)'	35E, 48S	–
(3) Silurian and Devonian rocks, Britain [10,16]						
22	Andesite Intrusions, Shelve Inlier		$<C_u$, $>LV_u$	M(a, t)'''	64E, 14N	7/13
23	Corndon Hill Phacolith		<entry 22	R(a, t)'''	69E, 15N	7/12
24	Dolerite Intrusions, Shelve Inlier		LV_{l-m}	R(a, t)'''	54E, 15N	6/11
25	E-W Dolerite Dykes, Shelve-Longmynd		LV_{l-m}	N(a, t)''''	230E, 16S	3/ 7
26	Moel-y-Golfa Andesite [8]		<C, >WE	N(t)'	242E, 16S	–
27	Breidden Hill Dolerite [9]		$<LV_u$, >WE	R(a, t)'''	346E, 43N	13/20
28	Lavas, Somerset and Gloucestershire [10]		WE	M(a, f)''	305E, 7N	12/18
29	Lorne Plateau Lavas	421 *	–	N(a, f)''''	141E, 2S	5/ 8
30	Garabal Hill–Glen Fyne Intrusion	404 *,**	–	N(a, t)'''	146E, 5N	18/30
31	Arrochar Complex	418 **		R(a, t)'''	324E, 8N	4/ 6
32	Lower O.R.S. Lavas and Sediments, Midland Valley [11]	394 **	<DT, S_u-D_l	M(a, t, f)''''	140E, 4N	4/ 7
33	Lower O.R.S. Lavas, Cheviot	390 **	D_l	M(a)'''	140E, 11N	12/17
34	Lower O.R.S. Lavas, Glencoe		S_u-D_l	N(a)'''	149E, 6S	18/15
35	Cautley Fell Dolerite		<LV, >CB	M(a)''	307E, 9S	–
36	Fishguard Volcanic Group (secondary magnetisation)		–	N(a)'''	119E, 2N	16/28
37	Shap Granite [12]	393 *,**	D	R(a, t)''	313E, 33S	4/ 7
38	SW-NE Minette Dykes, Cross Fell [13]		<LV, >CB	R(a)'''	316E, 23S	13/25
39	Middle O.R.S. Sediments, Foyers [14]		$\sim D_m$	R(a, t)''	327E, 30S	14/28
40	Hendre Dolerite, Berwyns		D	R(a, t)''	346E, 32S	–
41	Blodwell Keratophyre, Berwyns		D	R(a, t)''	349E, 33S	–
42	Upper O.R.S., Jedburgh		D_u	R–''	349E, 23S	5/ 9
43	Upper O.R.S., Bristol District		D_u-CB	R(a, f)'''	338E, 32S	5/10
44	Lavas, Orkney [15]	370 **	D_m	R(t)''	330E, 24S	–
45	Caithness Sandstone [15]		D_m	N(t)''	147E, 21N	–
46	Skomer Volcanic Group (Hercynian secondary magnetisation)		D_u-CB	N(a)'''	154E, 32N	9/17

Miscellaneous results from a.f. and/or thermally cleaned material, known or believed to fall in this time interval are Embleton diorite near Cockermouth (16°E, 14°N; [11]), Snead phacolith, Shelve Inlier (LV_m, 46°E, 18°N; [19]), NW-SE dolerite dyke swarm, S.W. Anglesey (130°E, 7°N; [64]).

Results not included in previous compilations [17] are: poles 2, 3 and 5 [20], poles 8, 13, 22–25, 40, 41 [19], poles 10 and 11 [10], poles 15 and 16 [14], poles 17, 18, 19, 37, 38 [11], pole 28 [22], pole 29 [21], pole 33 [23] and pole 45 [67].

* Rb-Sr age (calculated for ^{87}Rb half-life 4.85×10^{10} years), ** K-Ar age, *** fission track age on rock unit of palaeomagnetic study or on stratigraphically-related units.

Stratigraphic series are abbreviated as follows: AR = Arenig; LN = Llanvirn; LD = Llandeilo; C = Caradoc; AS = Ashgill; LV = Llandovery; WE = Wenlock; LU = Ludlow; DT = Downtonian; O.R.S. = Old Red Sandstone; l, m, u = lower, middle and upper divisions of the stratigraphic series.

O = Ordovician, S = Silurian, D = Devonian, CB = Carboniferous.

a = a.f. demagnetisation, t = thermal demagnetisation, f = fold test. The class of the palaeomagnetic pole determination is categorised as follows: '''' = determination based on 12 or more sites and more than 50 samples; ''' = determination based on 5–12 sites; '' = determination based on less than 5 sites but with a sample of magnetisations that is considered by authors to have averaged components of secular variation; ' = determination based on unit representing a single time event; these determinations can be considered as virtual geomagnetic poles only. *Dp* and *dm* are the semi-axes of the oval of confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. Confidence ovals are not given where the interval of magnetisation of the rock unit cannot be considered to have averaged secular variation.

Itemised notes:

[1] Original results recalculated after Kneen [59] and Morris [42] to include 12 sites in plutonic igneous rocks excluding country rocks.

[2] Mean of all 35 sites [9–11] yielding directions of magnetisation comparable to other British Ordovician data; all these sites are appreciably below entry 12 in the Borrowdale succession.

[3] Piper et al. [11] reported palaeomagnetic results for the Haweswater dolerite, a ca. 2-km^2 body intruding the Borrowdale Volcanic Series at 357.1°E, 54.6°N, but the tectonic significance of this result was uncertain. This body is, however, known to be closely genetically linked to the volcanics and should be corrected for the regional tilt (Dr. M.J.C. Nutt, personal communication, 1978); the resultant direction of magnetisation ($D = 315°$, $I = -60°$; [11]) is closer to directions found in the lower part of the succession [9,10] than in the upper part of the sequence (this paper). It is not included in this table because the dolerite body is affected by regional shear faulting which may have influenced the direction of magnetisation.

[4] This unit is Caradocian (Actonian) in age and separated from entry 12 by unconformity and a thin sedimentary suite (Fig. 1).

[5] Mean given by Thomas and Briden [14] for 52 sites in microgranite and grandodiorite intrusions of the North Wales igneous province.

[6] The mean direction of minette lamprophyre dykes from Northern England including sites from Ingleton and Stockdale (reversed) of this study is $D = 212°$, $I = 14°$, ($N = 11$ sites, $\alpha_{95} = 21°$). However, the site declinations range from 243° to 171° and inclinations range from +40° to −25°, and the virtual geomagnetic poles lie along the trend C → D in Fig. 4; is thus probable that the intrusion of these lamprophyres spanned a considerable period of Devonian times and that they are linked to the widespread granitic magmatism of this general age in Northern England.

[7] Igneous rocks of the Lower Palaeozoic Cautley Inlier, Howgill Fells (2.5°W, 54.3°N) comprise a rhyolite and intrusive felsite cut by later dolerites (Table 1).

[8] Tilt corrected mean after thermal demagnetisation is $D = 281°$, $I = -45°$ ($N = 13$, $R = 12.9$, $k = 98$, $\alpha_{95} = 4.2$); this displaces an earlier result based on a.f. demagnetisation only.

[9] Result is quoted prior to tilt correction of intruded sediments (see Piper [19]).

[10] The direction of magnetisation represented by group C poles (28–36) was first established by the work of Chamalaun and Creer [62] on Old Red Sandstone sediments of the Anglo Welsh area and Stubbs [63], some results included in calculation of result 32 on Scottish Devonian lavas.

[11] Mean of many results from several studies quoted in Briden et al. [17].

[12] The pole for the Carrock Fell Gabbro is listed in Morris [9] and Faller et al. [10] after correction for the tilt of Carboniferous sediments exposed 2–3 km to the east. If this correction is applied to poles 10B and 37 of this study it moves the pole from the Round Knott Dolerite to 358°E, 17°S and the pole from the Shap Granite to 213°E, 28°S.

[13] The mean direction of magnetisation of kersantite dykes from northern England including results of this study is $D = 242°$, $I = 86°$ ($N = 6$ sites, $\alpha_{95} = 12°$) and the palaeomagnetic pole is at 348°E, 51°N. These dykes are believed to be upper Ordovician in age and related to widespread granodiorite intrusion at depth [11].

[14] Calculated from sites in sandstones listed in Kneen [59].

[15] The bulk of these collections were rejected by authors (see also Briden et al. [17]).

[16] Extensive demagnetisation studies of sediments of this age (no group means given) are also reported by Turner and Archer [60] and Tarling et al. [61].

lion years. Sudden changes in the directions of apparent polar movements have been related to tectonic episodes in Precambrian times [30] and the detailed definition of Ordovician to Devonian polar movements permits this concept to be tested during the interval of the Caledonian orogeny. The first, second and third shifts coincide with low-strain folding episodes in the Caledonides [26]. The late Ordovician/early Silurion folding is widely developed in the paratectonic Caledonides, although in some areas deposition of sediments continued without break and in others the tectonic events cannot be separated with the precision necessary to isolate the effects of shifts $A \rightarrow B$ and $B \rightarrow C$. The shift $C \rightarrow D$ coincides with the widespread folding of Lower Old Red Sandstone rocks prior to deposition of Middle and Upper Old Red Sandstone rocks [26]. Conversely, the high-strain late Silurian folding appears to be entirely unrepresented by polar shift (it is also noteworthy that the Taconic and Acadian tectonic episodes in the Appalachian belt have no signature in the present a.p.w. record for North America – see Fig. 8). Thus it does not appear to be possible to correlate precisely the movements of this crustal unit with the history of deformation of the Caledonides, and it is implicit in this observation that at least these high strain episodes must be linked to some factor other than the large plate translations.

4. Anomalous geomagnetic fields

Poles 14–19 (at least in part of Upper Ordovician age) yield directions of magnetisation and pole positions (Fig. 7) removed from all other Palaeozoic studies. They are present as the predominant magnetisation in hypabyssal intrusions of the North Wales region distributed over 60 km [14] and plutons and dyke swarms of the Northern Lake District/Cross Fell region (Figs. 1 and 7) distributed over 40 km [11]; the regional distribution is along some 200 km of the southeast margin of the Iapetus suture (Fig. 7A). A regional tectonic explanation can probably be ruled out because (in addition to the arguments given by Thomas and Briden [14]) the Middle Caradocian Stockdale Rhyolite (Table 1) of the Lake District has a magnetisation close to the Cader Idris basalts of comparable age in North Wales (pole 15, Table 2).

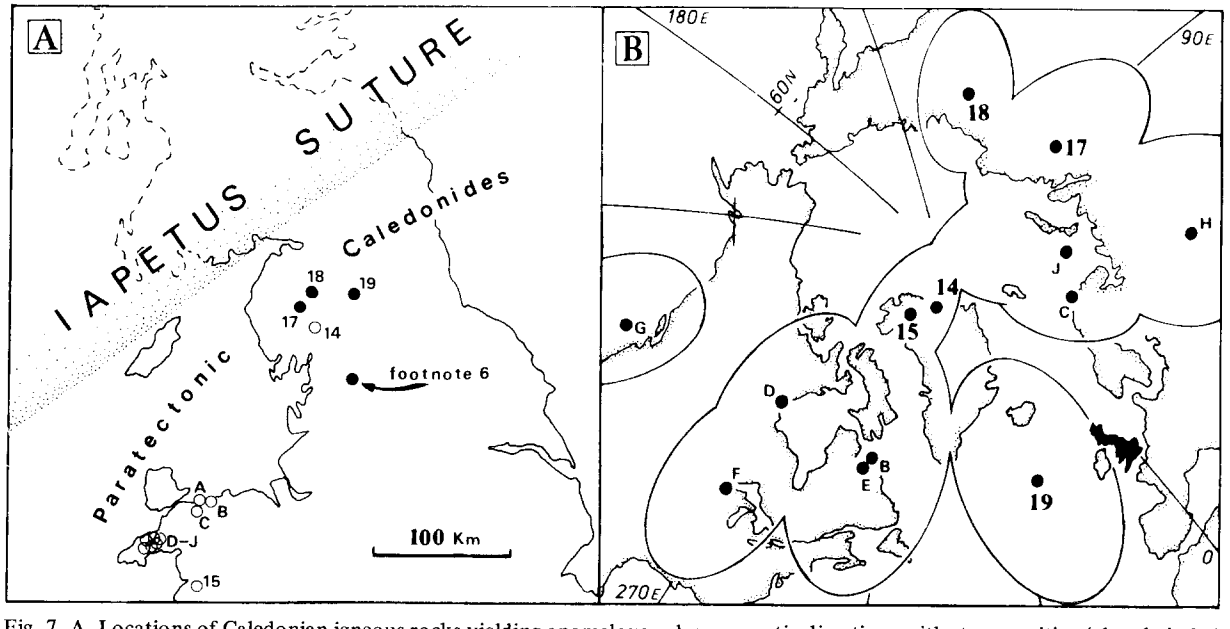


Fig. 7. A. Locations of Caledonian igneous rocks yielding anomalous palaeomagnetic directions with steep positive (closed circles) or negative (open circles) inclinations. B. Palaeomagnetic poles with 95% confidence ovals from these anomalous results; the numbers refer to poles listed in Table 2 and letters refer to hypabyssal intrusions of the North Wales igneous province listed in order of Thomas and Briden [14, table 1], for which the mean direction is listed as pole 16 (Table 2).

Large-scale translations can also be excluded firstly because latitudinal movements of Britain two orders of magnitude greater than any known plate movement would be required, and secondly because results 20 and 21 show that the pole had returned close to its typical Lower Ordovician position by Ashgill times.

The widespread intrusive rocks of northern England (poles 17–19, Table 2 yield steep positive directions of magnetisation while the anomalous intrusive bodies from North Wales (pole 16, Table 2) yield steep negative directions of magnetisation [14]. It is thus by no means certain that we are dealing here with the same period of anomalous field behaviour. However, the Threlkeld–St. John's microgranodiorite with steep positive magnetisation is also known from Rb-Sr studies to be approximately Caradocian in age [31], the known age of the Stockdale Rhyolite and the Cader Idris basalts both with the steep negative magnetisations. On the basis of this evidence, the period or periods of anomalous geomagnetic field behaviour are attributed to the Caradocian stage of Ordovician times. However, they cannot have occupied the whole of this interval because Lower Caradocian age tuffs from the Welsh Borderlands (pole 13, Table 2) and the Bail Hill Volcanics of the Southern Uplands (pole 5) yield mean directions close to pre-Caradocian Ordovician rocks [17]. The data documented in section 2 further show that these anomalous fields *commenced* after eruption of the last Borrowdale Volcanics (possibly late Llandeilo times) and prior to eruption of the Stockdale Rhyolite in Middle Caradocian times.

Palaeomagnetic results from Upper Ordovician rocks from elsewhere are needed to evaluate the geomagnetic significance of these data, but because they show that the geomagnetic field was not simply dipolar throughout the Lower Palaeozoic interval, they restrict the certainty with which we can draw tectonic conclusions from the remaining data.

5. Palaeomagnetic comparison with Continental Europe and North America

Palaeomagnetic poles from Continental Europe (including the Baltic Shield and Russian Platform) north of the Hercynian suture and east of the Uralian suture exhibit slight a.p.w. movement (Fig. 8A) from Ordovician to Devonian times, with none of the complexities of the British data from which they only become indistinguishable in Middle Devonian times. This comparison implies that the British region was part of a separate sub-plate until this time [17], and predicts that the North Sea Basin developed along a tectonic break defining a pre-Middle Devonian plate margin; it is confirmed by Upper Proterozoic palaeomagnetic results from the Precambrian basement (the

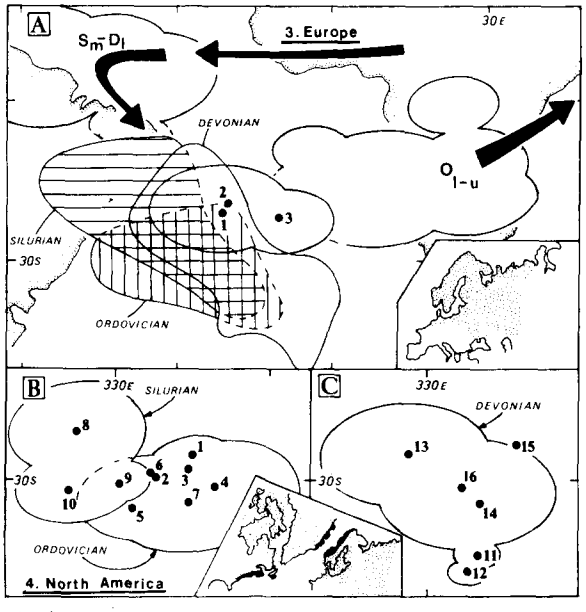


Fig. 8. A. The fields occupied by (9) Ordovician, (11) Silurian and (23) Devonian palaeomagnetic poles from the continental European plate [65,66] together with poles and confidence ovals for the Ringerike Sandstone (1, S_u), the Koragen Sandstone (2, D_l) and the Kvamshesten Formation (3, D_m) of Norway. B, C. Ordovician and Silurian poles (B) and Devonian poles (C) and confidence ovals from North America rotated to close the North Atlantic according to the Pangaean reconstruction of Bullard et al. [38]. The North American poles [37,66] with assigned ages in brackets are: 1 = Newfoundland Zinc Ores (O_l); 2 = St. George Limestone (O_l); 3 = St. George Dolostone (O_l); 4 = Moreton's Harbour basalt ($O_{l\text{-}m}$); 5 = Trenton Limestone (O_m); 6 = Juanita Formation (O_u); 7 = Beemerville Complex (O_u; 435 m.y.); 8 = Castanea Formation (S); 9 = Bloomsburg Formation (S_u); 10 = Clam Bank Group (D_l); 11 = Columbus Limestone (D_m); 12 = Delaware Limestone (D_m); 13 = Perry Formation Volcanics (D_u); 14 = Perry Formation Sediments (D_u); 15 = Perry Lavas (D_u); 16 = Perry Formation (D_u). The insets illustrate the continental blocks and/or reconstructions to which data apply.

"Midland Craton", see Fig. 5) of England and Wales which show no correlation with contemporaneous palaeomagnetic data from the Baltic Shield [32]. A confluence of Britain and Continental Europe in Devonian times along a zone running through the present North Sea area provides one explanation for the Caledonian Charnoid structural trend (NNW) of eastern England [17] which contrasts with the Caledonian structural trend (NE) in Scotland and the Anglo-Welsh area [33]. It is also consistent with the presence of Middle Devonian sediments as the oldest formations yet located in the axial region of the North Sea [34].

The Ordovician to Devonian North American palaeomagnetic poles also show only slight a.p.w. movement during this interval (Fig. 8) and although this has been explained in terms of widespread remagnetisation in Old Red Sandstone times [35], the primary nature of most of these magnetisations has since been convincingly demonstrated [36,37]. The North American data are indistinguishable from European data on the Pangaean reconstruction [38, 39] of the North Atlantic continents (cf. Fig. 8A–C); this agreement precludes any major post-Arenig strike-slip movement along the Arctic Caledonides, but since this belt is oriented approximately along a line of palaeolongitude, it cannot exclude some closure between the margins. A proximal relationship is indirectly suggested by reconstructions based on Late Precambrian palaeomagnetic data [40,41].

It follows from Figs. 4 and 8 that the British region between the orthotectonic Caledonides and the Hercynian Belt was also not coupled to the North American plate on the Pangaean reconstruction until Lower/Middle Devonian times. If we compare the mean palaeolatitudes of the British region as determined from groups *A*, *B*, and *C* data (Fig. 4) with the palaeolatitudes of Continental Europe/North America we observe (Fig. 9) that there is a systematic discrepancy in mean palaeolatitude between the two regions (positions 1, 2 and 3 in Fig. 9 see also Morris [42]) which persits whether we consider Ordovician, Lower

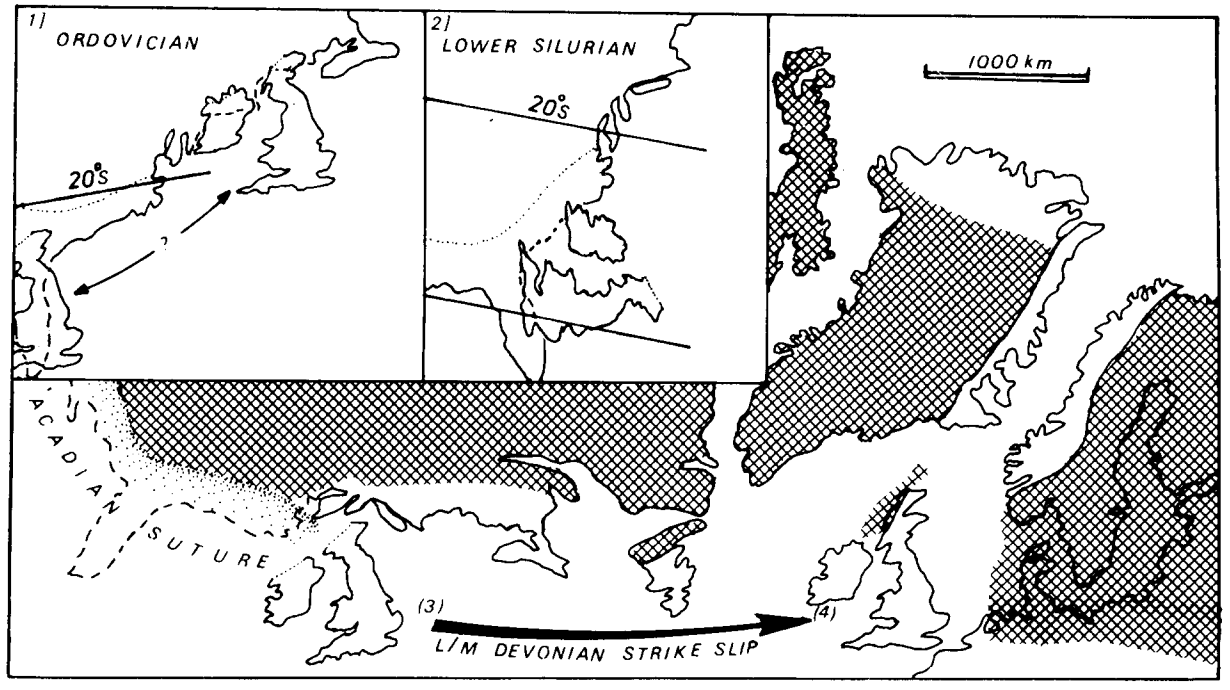


Fig. 9. Palaeogeographic reconstructions of Britain from Ordovician to Devonian times (positions 1–4) derived from groups of poles *A-D* compared with the Pangaean reconstruction of Bullard et al. [38] and obtained from palaeolatitudes inferred from the palaeomagnetic data of Figs. 4 and 7; mean palaeolatitude lines for North America are also illustrated for comparison with the inset diagrams of Fig. 4; areas of Precambrian cratonic basement are shaded.

Silurian, or Upper Silurian/Lower Devonian data; it does however disappear in Middle/Upper Devonian times ([38,39], and Fig. 9). The individual differences of orientation in Fig. 9 (based, notably in the case of the Lower Silurian results, on more restricted data sets) can be accommodated merely by local rotation. The palaeomagnetic data thus predict a major strike-slip motion with a mean value of 3600 km along or close to the line of orthotectonic Caledonides. The new results discussed in section 3 now restrict this possible motion to no more than a few million years close to the Lower/Middle Devonian boundary; this time interval is considerably less than the previous estimate of 20 m.y. [42] and the implied motion is greater. Also it is now clear that it can apply to the British region alone (Fig. 8) and does not pertain to Continental Europe. The picture that emerges is thus one of minimal post-Arenig separation across the Iapetus suture but of major strike-slip dislocation at late stages in the orogeny [42] and to this extent the Caledonian belt has analogies in the Alpine-Himalayan belts where zones of plate closure defined by ophiolitic rocks within fold belts have been succeeded after collision by large-scale transcurrent movements along plate boundaries separating continental blocks. It remains to test the geological implications of this comparison and to evaluate the palaeomagnetic grounds on which it is based.

6. Geological evidence and limitations of the palaeomagnetic data

The palaeolatitudinal discrepancy in Fig. 9 receives support from palaeontological data. Skevington [43] has shown that certain aspects of the graptolite faunas in Ordovician times defining the "Atlantic" and "Pacific" provinces reflect a strong climatic zonation with the only anomaly [43, Fig. 1] being associated with the British region in the Pangaean reconstruction; this anomaly is removed if this region lay 10° further to the south in Ordovician times.

Two possibilities could account for the large sinistral motion: either it might have taken place along some unrecognised lineament close to, or coincident with, the present continental margin, or it might have taken place along the Great Glen and subsidiary faults cutting the orthotectonic Caledonides (Fig. 5). The first possibility is unsatisfactory only in so far as it is difficult to test, but in many areas plate rupture has taken place along Precambrian and Palaeozoic lineaments (e.g. [44]). The second possibility requires motions an order of magnitude greater than the (also sinistral) movements currently recognised on the Great Glen fault in Scotland (Fig. 5, [45,46]) and the Leck-Leannon fault in Ireland [6]. Whilst this would require that such features as the Grampian metamorphic isograds [46] and the Dalradian succession are not to be directly correlated with features in the northern Highlands, it is not incompatible with the evidence for several reasons. Firstly, the Moinian succession north of the Great Glen area lies on Lewisian and Grenville age basement [47] and experienced an orogenic episode (the "Morarian") in late Precambrian times (ca. 730 m.y. [48]); south of the fault the supposed correlatives of the Moinian pass, probably conformably, into the Dalradian sequence with deposition continuing into Ordovician times. Secondly, the style of Caledonian deformation on either side of the fault zone is different [49] and probably took place at different intervals of the Arenig stage [50]; also slices of Precambrian basement are not observed in the Grampians and the strike of K-Ar uplift "chrontours" [50] is quite different on either side of the fault zone. Thirdly, the Dalradian sequence represents material derived from a southerly quadrant (summary of evidence by Phillips et al. [6]) unrelated to the Morarian episode. However, it is clear that positive geological evidence for a transcurrent movement of the magnitude implied by the palaeomagnetic evidence is lacking and it is more likely that only a portion of the total movement was accomodated by motion along this fault. Ultimately it will only be possible to decide between these possibilities from palaeomagnetic studies of areas north of the Great Glen fault. The current Precambrian palaeomagnetic data from the Lewisian foreland of northwest Scotland appear to agree approximately with North American data on the Pangaean reconstruction (e.g. [51]) but their age control is too poor to be sure of this.

As noted by Morris [42], this palaeomagnetic discrepancy also implies that Ordovician ophiolite sequences of the British Isles and North America are not to be directly correlated with one another along the strike on the Pangaean reconstruction [4]. Since

palaeomagnetic data now show that the western platform and possibly part of the central mobile belt of Newfoundland developed essentially in their present position with respect to North America [37], they would have been bypassed by the British region during the implied Devonian transcurrent movements. Consistent with such a lateral movement, the Newfoundland ophiolites were emplaced and deformed in Middle Devonian times [52], while the Atlantic margin of North America from Newfoundland to at least as far south as Maine was involved in Middle Devonian folding attributed to the Acadian orogeny [53]. The continuity of fish faunas between Gondwanaland and North America suggests that physical contact between the two plates was achieved in Middle Devonian times and McKerrow and Ziegler [54] argue for Middle Devonian orogeny in northwest South America although there is a disagreement as to which margin of Gondwanaland was involved in this collision [54,55]. In either case, in both time and place this collision would be appropriate to displacing the British sub-plate along the Appalachian margin until it became effectively locked between the North American and Continental European plates (Fig. 9).

There are, however, a number of limitations to the palaeomagnetic data which prevent unqualified acceptance of this tectonic model. They relate to the complexities of the Palaeozoic geomagnetic field evident from a number of the studies listed in Table 2. They leave some doubt both as to whether we should regard Palaeozoic a.p.w. movements for the British region simply in the context of the movement $A \rightarrow B \rightarrow C \rightarrow D$ illustrated in Fig. 4, and as to whether the data for Continental Europe/North America are even yet adequate to make the comparison on Fig. 8. These difficulties may be summarised under three headings:

(1) None of the studies yielding the anomalous results 14–19 (Table 2) have defined the dipole field recognised from other Ordovician studies, and their geomagnetic implications are, at present, quite uncertain. Also, although poles 14 and 15 are known to be of Upper Ordovician (Caradocian) age and pole 17 has a comparable radiometric age, the remaining anomalous poles can, from their geological relationships, only be defined as post-Lower Ordovician and (poles 18 and 19) pre-Carboniferous; it is possible that they relate to other parts of the interval under discussion.

(2) Two of the Upper Ordovician poles (Caradocian pole 5, and Ashgillian pole 21) fall between 10° and 20° south of the remainder of the British Ordovician data. There is no palaeolatitudinal discrepancy between these data and the Pangaean data of Fig. 7; against this observation we note that the Ordovician data from Ireland yield closely similar palaeolatitudes to the British data.

(3) The most detailed palaeomagnetic study yet made of Lower Devonian rocks [56] has shown that the geomagnetic field also had a complex character during at least part of this period; only 34 of 61 sites in lavas of the Strathmore region of the Midland Valley of Scotland yielded normal and reversely magnetised (N, R) directions defining a dipole field axis. Eighteen of the remaining sites defined an intermediate A (11 sites) direction and B (7 sites) direction. Further study by Kono [57] has confirmed this analysis; he finds no significant difference in palaeointensity estimates of the N, R, A and B field directions and argues for significant multipole field components in addition to dipole field components at this time.

Fifteen percent of the studies in Table 2 and a slightly higher percentage of the total data show that the geomagnetic field was not a simple dipolar field throughout this period of time. Conversely it can well be argued that the dipole field is now so well defined that the tectonic conclusions are unlikely to be far wrong; thus the a.p.w. path as defined in Fig. 4 is now based on a very large body of data originating from several different laboratories, and there is a general agreement on the timing of this path. Independent corroborative evidence is also forthcoming from the eastern seaboard of North America, where a comparable discrepancy is present between rocks of the eastern Appalachians and the remainder of North America [58]; the implied sinistral motion is also comparable although there is evidence that the anomaly here continues into Upper Devonian times. However, until the time significance of the geomagnetic anomalies is known, and until their significance in the Palaeozoic record of North America and Continental Europe is adequately investigated, the tectonic implications of Caledonian palaeomagnetic data as defined here and elsewhere cannot be regarded as entirely unequivocal.

Acknowledgements

Palaeomagnetic studies at Liverpool are supported by Natural Environment Research Council Grant GR/3/2398; I am grateful to Dr. W.A. Morris for useful discussions and reviewers for their valuable comments.

References

1 J.T. Wilson, Did the Atlantic close and then reopen?, Nature 211 (1966) 676–681.
2 W.B. Harland and R.A. Gayer, The Arctic Caledonides and earlier oceans, Geol. Mag. 109 (1972) 289–314.
3 W.Q. Kennedy, The Great Glen Fault, Q. J. Geol. Soc. London 102 (1946) 41–76.
4 J.F. Dewey, Evolution of the Appalachian/Caledonian orogen, Nature, London 222 (1969) 124–129.
5 A.H.G. Mitchell and W.S. McKerrow, Analogous evolution of the Burma orogen and the Scottish Caledonides, Geol. Soc. Am. Bull. 86 (1975) 305–315.
6 W.E.A. Phillips, C.J. Stillman and T. Murphy, A Caledonian plate tectonic model, J. Geol. Soc. London 132 (1976) 579–609.
7 J.F. Dewey and J.M. Bird, Mountain belts and the new global tectonics, J. Geophys. Res. 75 (1970) 2625–2647.
8 N.J. Soper and N.M.S. Numan, Structure and stratigraphy of the Borrowdale volcanic rocks of the Kentmere area, English Lake District, Geol. J. 9 (1974) 147–166.
9 W.A. Morris, Palaeomagnetic studies in the British Caledonides, Ph.D. Thesis, The Open University (1973).
10 A.M. Faller, J.C. Briden and W.A. Morris, Palaeomagnetic results from the Borrowdale Volcanic Group, English Lake District, Geophys. J.R. Astron. Soc. 48 (1977) 111–121.
11 J.D.A. Piper, A.S. McCook, K.P. Watkins, G.C. Brown and W.A. Morris, Palaeomagnetism and chronology of Caledonian igneous episodes in the Cross Fell Inlier and Northern Lake District, Liverpool Manchester Geol. J. 13 (1978) 73–92.
12 G.H. Mitchell, The succession and structure of the Borrowdale Volcanic Series in Troutbeck, Kentmere and the western part of Long Sleddale (Westmoreland), Q. J. Geol. Soc. London 85 (1929) 9–42.
13 M.W. McElhinny, Statistical significance of the fold test in palaeomagnetism, Geophys. J. R. Astron. Soc. 8 (1964) 338–340.
14 C. Thomas and J.C. Briden, Anomalous geomagnetic field during the late Ordovician, Nature 259 (1976) 380–382.
15 J.E. Marr, The Lower Palaeozoic rocks of the Cautley District (Yorkshire), Q. J. Geol. Soc. London 69 (1913) 1–18.
16 W. Edwards, F.M. Trotter and D.A. Wray, British Regional Geology – The Pennines and Adjacent Areas (H.M. Stationery Office, London, 1954) 86 pp.
17 J.C. Briden, W.A. Morris and J.D.A. Piper, Palaeomagnetic studies in the British Caledonides, VI. Regional and global implications, Geophys. J. R. Astron. Soc. 34 (1973) 108–146.
18 W.B.R. King and W.H. Wilcockson, The Lower Palaeozoic rocks of Austwick and Horton-in-Ribblesdale, Yorkshire, Q. J. Geol. Soc. London 90 (1934) 7–31.
19 J.D.A. Piper, Palaeomagnetic survey of the (Palaeozoic) Shelve Inlier and Berwyn Hills, Welsh Borderlands, Geophys. J. R. Astron. Soc. 53 (1978) 355–371.
20 J.D.A. Piper, Palaeomagnetism and palaeogeography of the Southern Uplands Block, in Ordovician times, Scott. J. Geol. 14 (1978) 93–107.
21 A.G. Latham and J.C. Briden, Palaeomagnetic field directions in Siluro-Devonian lavas of the Lorne Plateau, Scotland, and their regional significance, Geophys. J. R. Astron. Soc. 43 (1975) 243–252.
22 J.D.A. Piper, Palaeomagnetism of Silurian lavas of Somerset and Gloucestershire, England, Earth Planet. Sci. Lett. 25 (1975) 355–360.
23 L. Thorning, Palaeomagnetic results from Lower Devonian rocks of the Cheviot Hills, North England, Geophys. J. R. Astron. Soc. 36 (1974) 487–496.
24 J.G. Mitchell, Potassium-argon ages from the Cheviot Hills, Northern England, Geol. Mag. 109 (1972) 421–427.
25 P.E. Brown, J.A. Miller and N.J. Soper, Age of the principal intrusions of the Lake District, Proc. Yorks. Geol. Soc. 34 (1964) 331–342.
26 J.F. Dewey, Structure and sequence in paratectonic British Caledonides, in: North Atlantic Geology and Continental Drift, A Symposium, M. Kay, ed., Am. Assoc. Pet. Geol. Mem. 12 (1969) 309–335.
27 W.B. Harland and E.H. Francis, eds., The Phanerozoic Time Scale, A Supplement, Geol. Soc. London, Spec. Publ. 5 (1976) 356 pp.
28 W.A. Morris, J.C. Briden, J.D.A. Piper and J.T. Sallomy, Palaeomagnetic studies in the British Caledonides, V. Miscellaneous new data, Geophys. J. R. Astron. Soc. 34 (1973) 69–107.
29 W.A. Morris, Palaeomagnetic results from the Lower Palaeozoic of Ireland, Can. J. Earth Sci. 13 (1976) 294–304.
30 J.A. Donaldson, E. Irving, J.C. McGlynn and J.K. Park, Drift of the Canadian Shield, in: Implications of Continental Drift to the Earth Sciences. D.H. Tarling and S.K. Runcorn, eds. (Academic Press, London, 1973) 3–18.
31 A.J. Wadge, R.R. Harding and D.P.F. Derbyshire, Rb/Sr age and field relations of the Threlkeld Microgranite, Proc. Yorks. Geol. Soc. 40 (976) 211–222.
32 J.D.A. Piper, A palaeomagnetic study of Late Precambrian rocks of the Midland Craton of England and Wales, Phys. Earth Planet. Inter. 19 (1979) 59–72.
33 J.S. Turner, The deeper structure of central and northern England, Proc. Yorks. Geol. Soc. 27 (1949) 280–297.
34 P.E. Kent, Review of North Sea Basin development, Q. J. Geol. Soc. London 131 (1975) 435–468.

35 K.M. Creer, Palaeozoic palaeomagnetism, Nature 219 (1968) 246–250.
36 M.W. McElhinny and N.D. Opdyke, Remagnetisation hypothesis discounted – a palaeomagnetic study of the Trenton Limestone, New York State, Geol. Soc. Am. Bull. 84 (1973) 3697–3708.
37 E.R. Deutsch and K.V. Rao, New palaeomagnetic evidence fails to support rotation of Western Newfoundland, Nature 266 (1977) 314–317.
38 E.C. Bullard, J.E. Everett and A.G. Smith, The fit of the continents around the Atlantic, Philos. Trans. R. Soc. London, Ser. A, 258 (1965) 41–75.
39 J.L. Roy, A pattern of rupture of the eastern North American-western European palaeoblock, Earth Planet. Sci. Lett. 14 (1972) 103–114.
40 H. Ueno, E. Irving and R.H. McNutt, Palaeomagnetism of the Whitestone anorthosite and diorite, the Grenville Polar Track and the relative motions of the Laurentian and Baltic Shields, Can. J. Earth Sci. 12 (1975) 209–226.
41 P.J. Patchett and G. Bylund, Age of Grenville Belt magnetisation: Rb-Sr and palaeomagnetic evidence from Swedish dolerites, Earth Planet. Sci. Lett. 35 (1977) 92–104.
42 W.A. Morris, Transcurrent motion determined palaeomagnetically in the northern Appalachians and Caledonides and the Acadian Orogeny, Can. J. Earth Sci. 13 (1976) 1236–1243.
43 D. Skevington, Controls influencing the composition and distribution of Ordovician graptolite faunal provinces, Palaeontology, Spec. Paper 13 (1974) 59–64.
44 R.B. McConnell, Geological development of the rift system of Eastern Africa, Geol. Soc. Am. Bull. 83 (1972) 2549–2572.
45 W.S. Pitcher, North East trending faults of Scotland and Ireland and chronology of displacements, in: North Atlantic Geology and Continental Drift, A Symposium, M. Kay, ed., Am. Assoc. Pet. Geol., Mem. 12 (1969) 724–733.
46 J.A. Winchester, Pattern of regional metamorphism suggests a sinistral displacement of 160 km along the Great Glen Fault, Nature, Phys. Sci. 246 (1973) 81–84.
47 M. Brook, M.S. Brewer and D. Powell, Grenville age for rocks in the Moine of north-western Scotland, Nature 260 (1976) 515–517.
48 O. van Breemen, R.T. Pidgeon and M.R.W. Johnson, Precambrian and Palaeozoic pegmatites in the Moines of northern Scotland, J. Geol. Soc. London 130 (1974) 493–507.
49 M.S. Garson and J. Plant, Alpine type ultramafic rocks and episodic mountain building in the Scottish Highlands, Nature 242 (1973) 34–38.
50 J.F. Dewey and R.J. Pankhurst, The evolution of the Scottish Caledonides in relation to their isotopic age pattern, Trans. R. Soc. Edinburgh 68 (1970) 361–389.
51 W.A. Morris and J.L. Roy, Discovery of the Hadrynian Polar Track and further study of the Grenville Problem, Nature 266 (1977) 689–692.
52 J.M. Bird, J.F. Dewey and W.S.F. Kidd, Proto-Atlantic oceanic crust and mantle: Appalachian/Caledonian ophiolites, Nature, Phys. Sci. 231 (1971) 28–31.
53 J. Rodgers, The Tectonics of the Appalachians (John Wiley, New York, N.Y., 1970).
54 W.S. McKerrow and A.M. Ziegler, Palaeozoic oceans, Nature, Phys. Sci. 240 (1972) 92–94.
55 A.G. Smith, J.C. Briden and G.E. Drewry, Phanerozoic world maps, Palaeontology, Spec. Paper 12 (1973) 1–42.
56 J.T. Sallomy and J.D.A. Piper, Palaeomagnetic studies in the British Caledonides, IV. Lower Devonian lavas of the Strathmore region, Scotland, Geophys. J. R. Astron. Soc. 34 (1973) 47–68.
57 M. Kono, Palaeomagnetism and palaeointensity studies of Scottish Devonian volcanic rocks, Geophys. J. R. Astron. Soc. 56 (1979) 385–396.
58 D.V. Kent and N.D. Opdyke, Devonian palaeomagnetism and the formation of the Euramerican continent, Trans. Am. Geophys. Union 59 (1978) 263.
59 S.J. Kneen, The palaeomagnetism of the Foyers Plutonic Complex, Inverness-shire, Geophys. J. R. Astron. Soc. 32 (1973) 52–63.
60 P. Turner and R. Archer, Magnetisation history of Lower Old Red Sandstones from the Gamrie Outlier, Scotland, Earth Planet. Sci. Lett. 27 (1975) 240–250.
61 D.H. Tarling, R.N. Donovan, J.M. Abou-Deeb and S.I. El-Batrouk, Palaeomagnetic dating of haematite genesis in Orcadian Basin sediments, Scott. J. Geol. 12 (1976) 125–134.
62 F.H. Chamalaun and K.M. Creer, Thermal demagnetisation studies on the Old Red Sandstone of the Anglo-Welsh cuvette, J. Geophys. Res. 69 (1964) 1607–1616.
63 P.H.S. Stubbs, Continental drift and polar wandering, a palaeomagnetic study of the British and European Trias and the British Old Red Sandstone, Ph.D. Thesis, University of London (1958) 158 pp.
64 J.D.A. Piper, Magnetic properties of Precambrian pillow lavas of the Mona Complex and a related dyke swarm, Anglesey, Wales, Liverpool Manchester Geol. J. 11 (1976) 189–201.
65 A. Hicken, E. Irving, L.K. Law and J. Hastie, Catalogue of Palaeomagnetic Directions and Poles. First Issue, Palaeozoic Results, Earth Phys. Branch Publ. 45, Ottawa, Ont. (1972).
66 E. Irving, E. Tanczyk and J. Hastie, Catalogue of Palaeomagnetic Directions and Poles. Third Issue, Palaeozoic Results, Earth Phys. Branch Publ. 75, Ottawa, Ont. (1976).
67 K.M. Storetvedt, A possible large-scale sinistral displacement along the Great Glen Fault in Scotland, Geol. Mag. 111 (1974) 23–30.